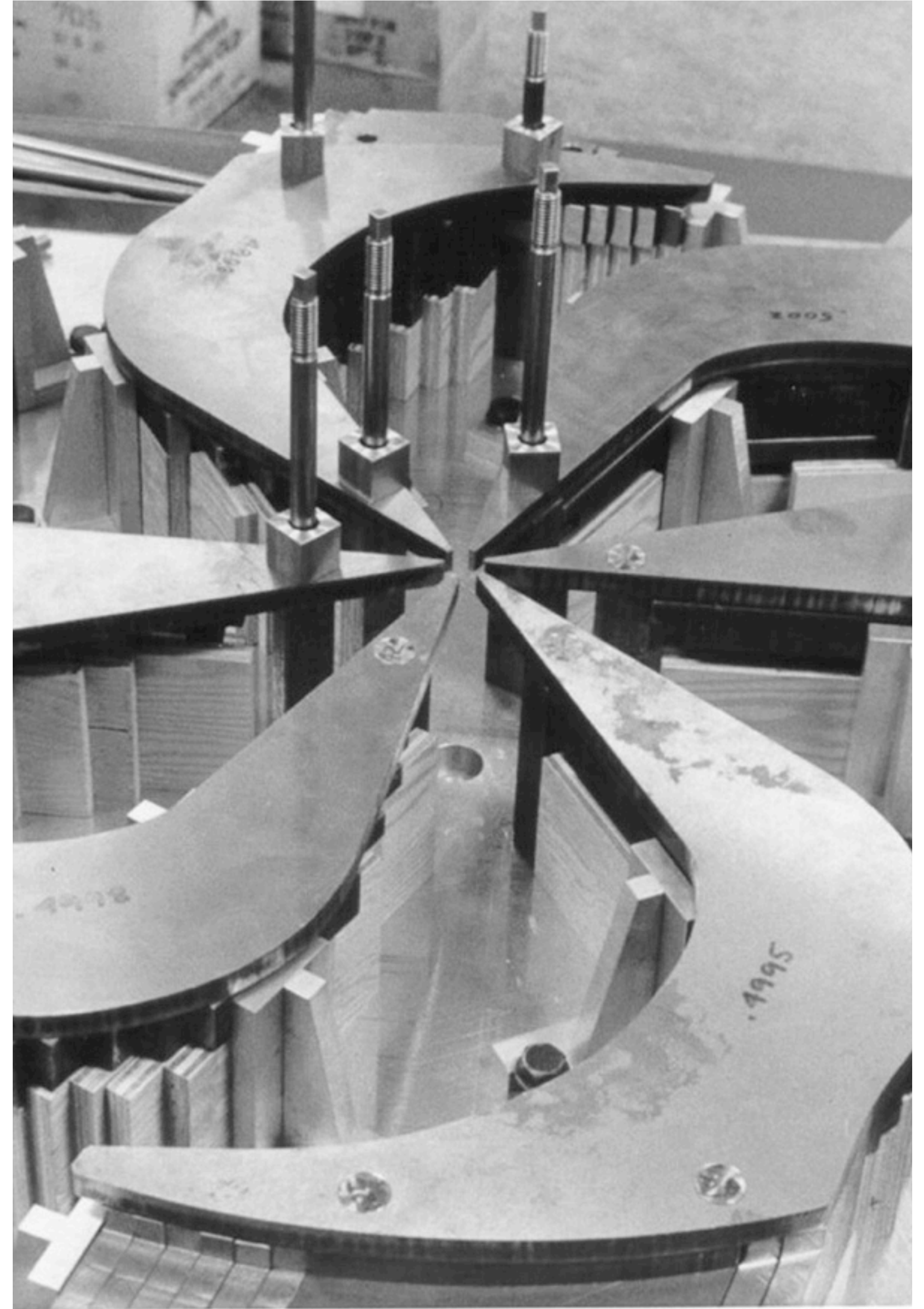


Summary of WorkFest work

Laura Miller

Software Meeting
2026-05-27



Full G4 sim work

2

- Updating G4 to be more reflective of reality so we can try to better model the beamline (and other environmental) backgrounds
- Supposed to be in parallel with Ethan's FLUKA sims, as a second cross check
- Done at workfest: added magnetic field for the optics downstream of the target chamber
- Still to do: add physical beam spool piece and PMQs

Relevant changes

- New flag added in G4 that can be set in the macro: `/det/doDSOptics true`
- Can be set in the `darklight_30MeV.mac` under the normal magnetic field call
- Setting to true will toggle the downstream optics on, otherwise defaults to false

